CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of tangents that can be drawn to a circle from a point lying inside it is: 1

(a) infinite **(b)** 2 **(c)** 1 **(d)** 0

2. A point P is 15 cm from the centre O of a circle of radius 9 cm. The length of the tangent drawn from P to the circle is: 1

(a) 24 cm **(b)** 12 cm **(c)** 144 cm **(d)** 6 cm

3. The length of the tangent from an external point P to a circle is 15 cm and P is 17 cm from the centre. The radius of the circle is: 1

(a) 2 cm **(b)** 32 cm **(c)** 16 cm **(d)** 8 cm

4. A tangent to a circle of radius 20 cm has length 21 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 41 cm **(b)** 58 cm **(c)** 1 cm **(d)** 29 cm

5. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 9 cm. The length of each tangent is 12 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 21 cm **(b)** 24 cm **(c)** 42 cm **(d)** 48 cm

6. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 140^\circ$. The $\angle APB$ is: 1

(a) 40° **(b)** 20° **(c)** 140° **(d)** 280°

7. PA is a tangent at A to a circle with centre O, and $\angle OPA = 55^\circ$. The $\angle AOP$ is: 1

(a) 125° **(b)** 145° **(c)** 35° **(d)** 55°

8. The number of tangents that can be drawn to a circle from a point lying on it is: 1

(a) 1 **(b)** 2 **(c)** 0 **(d)** infinite

9. A point P is 17 cm from the centre O of a circle of radius 8 cm. The length of the tangent drawn from P to the circle is: 1

(a) 9 cm **(b)** 15 cm **(c)** 225 cm **(d)** 25 cm

10. The length of the tangent from an external point P to a circle is 16 cm and P is 20 cm from the centre. The radius of the circle is: 1

(a) 12 cm **(b)** 4 cm **(c)** 36 cm **(d)** 17 cm

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 150°$. Find the $\angle APB$. 2

12. A point P is 30 cm from the centre O of a circle of radius 24 cm. Find the length of the tangent drawn from P to the circle. 2

13. A tangent to a circle of radius 12 cm has length 9 cm from the external point P to the point of contact. Find the distance of P from the centre of the circle. 2

14. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 80°$. Find the $\angle APB$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 20 cm. The length of each tangent is 21 cm. Find the perimeter of the quadrilateral $PAOB$. 3

16. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 8 cm. The length of each tangent is 15 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

17. PA is a tangent at A to a circle with centre O, and $\angle OPA = 15°$. Find the $\angle AOP$. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 18 cm. The length of each tangent is 24 cm. Find the perimeter of the quadrilateral $PAOB$. 3

18. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 12 cm. The length of each tangent is 16 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

OR

PA is a tangent at A to a circle with centre O, and $\angle OPA = 65°$. Find the $\angle AOP$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Tangents PA and PB are drawn from a point P at a distance of 17 cm from the centre O of a circle of radius 8 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 30 cm and 18 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

20. Tangents PA and PB are drawn from a point P at a distance of 20 cm from the centre O of a circle of radius 12 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

Two concentric circles have radii 10 cm and 8 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 5

END OF QUESTION PAPER — **Total: 40 Marks**